

LOS Theory Assignment 5

Command Line Arguments, File operators, Read and Filters

Write a Bash script that accepts a filename as the first command-line argument.

- Check whether an argument is provided.
- Check whether the file exists using a file operator.
- If the file exists, display the first 5 lines using head.
- Otherwise, display an appropriate error message.

filehead.sh:

```
#!/usr/bin/bash
```

```
if [ -z "$1" ]
```

```
then
```

```
    echo "Error: No filename provided."
```

```
    exit 1
```

```
fi
```

```
if [ -f "$1" ]
```

```
then
```

```
    echo "First 5 lines of the file:"
```

```
    head -n 5 "$1"
```

```
else
```

```
    echo "Error: File does not exist."
```

```
fi
```

File Type and Permission Checker

Write a Bash script that accepts a path as a command-line argument and determines whether it is:

- a regular file,
- a directory,
- readable,
- writable, and
- executable.

Display suitable messages using file operators such as `-f`, `-d`, `-r`, `-w`, and `-x`.

checkfile.sh

```
#!/usr/bin/bash
```

```
if [ -z "$1" ]
```

```
then
```

```
    echo "Error: No path provided."
```

```
    exit 1
```

```
fi
```

```
if [ -f "$1" ]
```

```
then
```

```
    echo "$1 is a regular file."
```

```
fi
```

```
if [ -d "$1" ]
```

```
then
```

```
    echo "$1 is a directory."
```

```
fi
```

```
if [ -r "$1" ]
```

```
then
```

```
    echo "$1 is readable."
```

```
else
```

```
    echo "$1 is not readable."
```

```
fi
```

```
if [ -w "$1" ]
```

```
then
```

```
    echo "$1 is writable."
```

```
else
```

```
    echo "$1 is not writable."
```

```
fi
if [ -x "$1" ]
then
    echo "$1 is executable."
else
    echo "$1 is not executable."
fi
```

Write a Bash script that accepts a log file as a command-line argument and searches for the words:

- **failed, denied, unauthorized, and error.**

```
searchlog.sh

#!/usr/bin/bash

if [ -z "$1" ]
then
    echo "Error: No log file provided."
    exit 1
fi

if [ ! -f "$1" ]
then
    echo "Error: Log file does not exist."
    exit 1
fi

grep -Ei "failed|denied|unauthorized|error" "$1"
```

Write a Bash script that accepts a filename and checks whether:

- the file exists,
- it is a regular file,
- it is empty or non-empty.
- If it is non-empty, display the first 5 lines.

filecheck.sh

```
#!/usr/bin/bash
```

```
if [ -z "$1" ]
```

```
then
```

```
    echo "Error: No filename provided."
```

```
    exit 1
```

```
fi
```

```
if [ -e "$1" ]
```

```
then
```

```
    echo "File exists."
```

```
else
```

```
    echo "File does not exist."
```

```
    exit 1
```

```
fi
```

```
if [ -f "$1" ]
```

```
then
```

```
    echo "It is a regular file."
```

```
else
```

```
    echo "It is not a regular file."
    exit 1
fi
if [ -s "$1" ]
then
    echo "File is non-empty."
    echo "First 5 lines:"
    head -n 5 "$1"
else
    echo "File is empty."
fi
```

Write a Bash script named `investigate.sh` that accepts a log file as a command-line argument.

- **The script must first verify that the file exists and is readable. Then ask the investigator to enter a search term.**
- **Display the first 5 matching records and report the total number of matching records.**

```
investigate.sh:
#!/usr/bin/bash

if [ -z "$1" ]
then
    echo "Error: No log file provided."
    exit 1
fi

if [ ! -e "$1" ]
```

```
then

    echo "Error: File does not exist."

    exit 1

fi

if [ ! -r "$1" ]

then

    echo "Error: File is not readable."

    exit 1

fi

read -p "Enter search term: " term

echo "First 5 matching records:"

grep -i "$term" "$1" | head -n 5

count=$(grep -ic "$term" "$1")

echo "Total matching records: $count"
```